

Introduction to Statistical Machine Learning

Kaggle Project: Covid-19 Awareness and Covid-19 Cases in Ohio

SeungHyun Nam and Vansh Brijesh Desai (Group Name: Vansh Nam)

AS&E, University of Rochester

Abstract. In this comprehensive study, the relationship between COVID-19 awareness, as measured through social media analytics in Ohio, and the pandemic's epidemiological impact, quantified by case and death counts alongside socioeconomic and demographic data, was scrutinized using a variety of machine learning models. Leveraging a dataset that amalgamates over 46 million tweets to gauge public sentiment and awareness with detailed county-level pandemic statistics, this research aimed to identify patterns that could potentially inform public health strategies. The exploration involved novel social media-derived metrics and conventional demographic indicators to understand the dynamics of public engagement with the pandemic. Among the models evaluated, the XG-Boost Regressor emerged as the most effective, demonstrating superior predictive performance on a test dataset that constituted 70% of the total observations, a reversal from traditional data splits in machine learning challenges. This key finding underscores the model's robustness in capturing intricate data patterns and highlights the critical importance of selecting appropriate machine learning techniques to forecast disease spread accurately. This study leverages advanced machine learning models to explore the complex dynamics of public engagement and information dissemination, setting the foundation for a detailed examination in the following sections.

1 Introduction

1.1 Problem Statement

The COVID-19 pandemic has presented an unprecedented challenge to public health systems globally, necessitating a multifaceted response strategy. An integral part of managing the pandemic's impact involves understanding public awareness and response to health advisories and guidelines. This study focuses on Ohio, a state that has experienced varied pandemic intensities and responses across its counties. By analyzing a dataset comprised of COVID-19 case counts, deaths, and a wealth of demographic information alongside measures of public awareness derived from social media data, this paper aims to uncover patterns and correlations that may inform future public health strategies. The significance of this study lies not only in its potential to aid in managing ongoing public health responses but also in its contribution to the broader discourse on the role of information dissemination and public engagement in health crises.

2 Data

2.1 Data Acquisition

The dataset used in this study provides a comprehensive overview of COVID-19's impact on Ohio's counties, coupled with an innovative measure of public awareness derived from social media activity. Specifically, the dataset includes daily case and death counts, alongside demographic variables such as population size, employment rates, and median household earnings. A notable feature of this dataset is the inclusion of "awareness" variables, quantified through the analysis of over 46 million tweets related to COVID-19, employing hashtags as a proxy for public interest and concern over various pandemic-related topics. These awareness variables are further processed through Jaccard, Cosine, and Intersection similarity measures to quantify the intensity of discussion around COVID-19. This multi-dimensional dataset, encompassing both traditional demographic information and novel social media-derived metrics, provides a unique lens through which the interaction between public awareness and pandemic outcomes can be explored. With a total of 3141 observations for training and 7331 for testing, the dataset offers a robust foundation for statistical analysis and machine learning modeling.

- **index**: Index associated with the observation (found only in the test set)
- **county**: Name of the Ohio county that is associated with the data point
- **cases**: Number of Covid-19 cases on a given day in a county of Ohio (found only in the training set)
- **deaths**: Number of Covid-19-related deaths on a given day in a county of Ohio
- **date index converted**: An anonymized index value for the day associated with awareness
- **county data length**: Number of tweets posted on a given day to calculate awareness level
- **total pop**: Population of the county (x 1000)
- **percent 25 34**: Percentage of the people who are between 25 and 34 years old
- **percent highschool**: Percentage of the people who have at least finished high school
- **labor force rate**: Percentage of the adult people who are currently employed
- **unemployment rate**: Percentage of unemployment in the county
- **median housing cost**: Median cost of a house in the county
- **median household earnings**: Median annual earnings of a household in the county
- **median worker earnings**: Median annual earnings of an employee in the county
- **percent insured**: Percentage of the people who currently have health insurance
- **percent married**: Percentage of the people who are currently married
- **poverty rate**: Percentage of the people who fall under the poverty line
- **median property value**: Median value of a property
- **percent white**: Percentage of white people in the county

2.2 Data EDA

The exploratory data analysis (EDA) for the COVID-19 awareness in Ohio provides a rich perspective on public engagement across the state’s counties, as well as the temporal evolution of discourse on various pandemic-related topics. The visualizations offered in the dataset highlight key findings which we will discuss below.

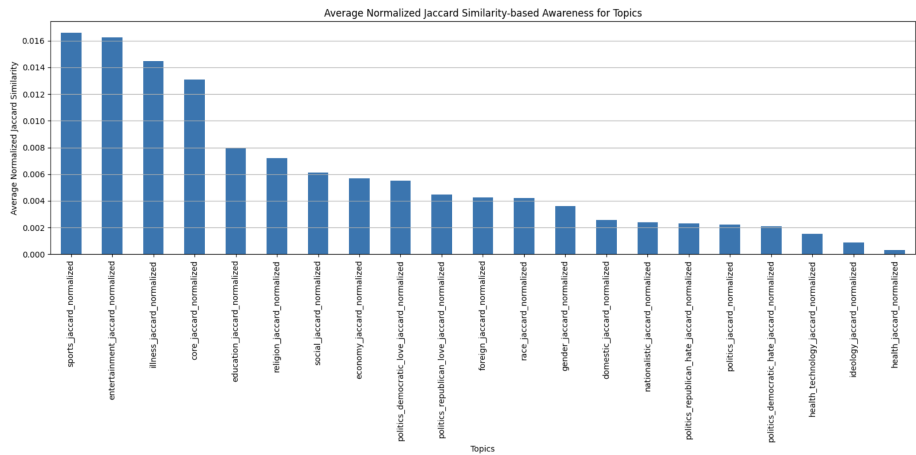


Fig. 1. Average Normalized Jaccard Similarity-based Awareness for Topics

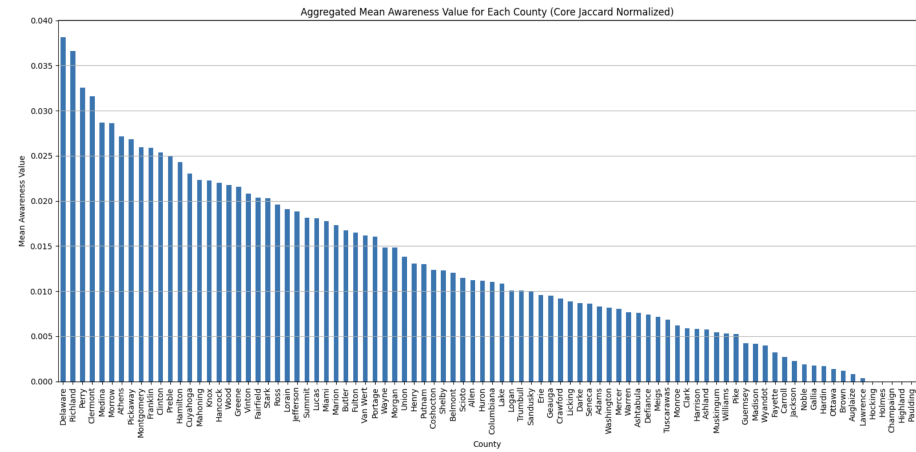


Fig. 2. Aggregated Mean Awareness Value for Each County (Core Jaccard Normalized)

For Figure 1, "Average Normalized Jaccard Similarity-based Awareness for Topics," the bar chart strategically ranks various topics by their average normalized Jaccard similarity-based awareness values, delineating from highest to

lowest. It distinctly reveals sports and entertainment as the primary focus of public interest, leading the awareness rankings. These sectors are closely followed by illness and core subjects related to general COVID-19 awareness, underscoring a significant public engagement with health-related themes. Moreover, topics such as education, religion, and social issues demonstrate notable average awareness levels, albeit slightly subdued compared to the frontrunners. In stark contrast, areas like health (excluding technology), ideology, and health technology are positioned at the lower echelons of the awareness scale, indicating a divergence in public interest towards these essential yet less attended to issues.

For Figure 2, "Aggregated Mean Awareness Value for Each County (Core Jaccard Normalized)," Delaware County is highlighted as the epitome of COVID-19 awareness across all analyzed counties, as measured by the core jaccard normalized index. This prominence underscores Delaware's superior level of public consciousness and understanding concerning COVID-19 topics relative to other counties in the dataset. The variation in awareness levels observed across different counties could be reflective of multiple contributory factors. These might encompass the intensity and reach of local COVID-19 informational campaigns, the tangible impact of the pandemic on the local populace, or the degree of access to and availability of informational resources and support related to COVID-19 within each county. Together, these insights from Figures 1 and 2 shed light on the landscape of public awareness regarding COVID-19, highlighting both the topics that captivate the most attention and the geographical variations in awareness levels.

Ohio Counties Colored by Average COVID-19 Cases per Capita



Fig. 3. Ohio Counties by Average COVID-19 Cases per Capita

Ohio Counties Colored by Average COVID-19 Deaths per Capita

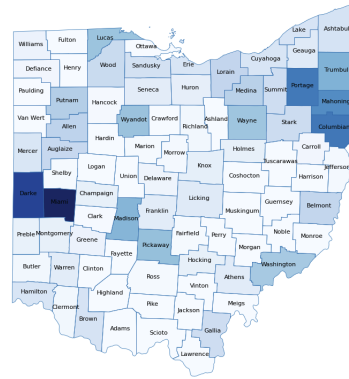


Fig. 4. Ohio Counties by Average COVID-19 Deaths per Capita

For Figure 3, representing "Ohio Counties Colored by Average COVID-19 Cases per Capita," the map delineates the geographical distribution of COVID-19 case rates across the counties. The varying shades of color intensity correlate

with the average number of cases per capita, with darker hues indicating higher case rates. The visualization suggests that there are specific counties, possibly those with denser populations or urban hubs, that experienced a higher prevalence of COVID-19 cases. This could be attributed to factors like community transmission dynamics, the concentration of testing facilities, or local response measures.

Turning to Figure 4, which depicts "Ohio Counties Colored by Average COVID-19 Deaths per Capita," a similar color gradient is employed to illustrate the death rates due to COVID-19. The map underscores that while some counties share a higher burden of cases, it does not uniformly translate to higher death rates, as seen by the variation in color intensity. Differences in health-care infrastructure, age demographics, or the presence of comorbidities within the population might inform this discrepancy. Overall, the map provides a stark visual of the pandemic's mortality impact at the county level, reflecting the unequal toll of the crisis across regions.

In summary, it emerges that the top 5 counties with the most pronounced number of per capita COVID-19 cases in Ohio are Pickaway, Marion, Lucas, Columbiana, and Mahoning. Additionally, when considering the gravest outcome of the pandemic—mortality rates per capita—the five counties that stand out are Miami, Darke, Columbiana, Portage, and Mahoning. This data underscores the disparities in the spread and impact of COVID-19 across the state, with certain regions bearing a disproportionately higher brunt of cases and fatalities. The insights drawn from this analysis can be pivotal for public health strategizing and resource allocation to address the needs of the most affected areas.

3 Method

3.1 Data Pre-Processing

Categorical Variable Conversion: The dataset contains two categorical variables: 'date_index_converted' and 'county'. For 'date_index_converted', we employ regular expressions to convert this variable into integers, facilitating numerical analysis. Regarding the 'county' variable, we assign a rank based on the number of COVID-19 cases, with the ranking reflecting the severity in descending order. This ranked data is then utilized in both training and testing datasets to maintain consistency and relevance. Although data has no null value but some of similarity columns are highly sparse. some column has negligible value which has range about $1e-6$.

3.2 Feature Reduction

Similarity Feature Analysis: We categorize an additional feature set based on similarity metrics such as Jaccard index, cosine similarity, and intersection. By calculating the principal components of these similarity features, we aim to

reduce dimensionality while retaining significant variance. The principal components are further processed through spectral embedding to visualize data grouping and relationship patterns, providing insights into the underlying structure. The spectral components are in form of curve which depicts that one component is in quadratic relationship with other components. Although this components are not helping to improve result of model.

Correlation Analysis: A correlation matrix is used to evaluate the relationships between principal components. However, due to high correlation, indicating redundancy, we decide against using these components for model improvement.

Demographic and Social Feature Integration: While converting demographic variables into categorical ones did not yield substantial accuracy improvements, incorporating social features (e.g., 'county_data_length', age demographics, labor force participation, unemployment rates, and median household earnings) significantly enhances model performance.

Data Transformation and Feature Selection: Attempts to apply a logarithmic transformation to the target variable (e.g., number of COVID-19 cases) to improve normality and reduce skewness did not prove effective, as indicated by high mean squared error (MSE) rates. An important preprocessing step involves filtering the dataset to exclude records before a certain 'date_index_converted' threshold (e.g., 80), based on the observation that the number of cases before this point is consistently zero. This helps focus the model training on more relevant data.

3.3 Test-Train Splitting

Data Splitting Strategy: The dataset is divided into training and testing sets using a stratified approach to ensure that both sets are representative of the overall data distribution, particularly concerning the target variable (COVID-19 cases). This approach improves the model's generalization ability to unseen data.

3.4 Model Selection

XGBoost Implementation: XGBoost stands out for regression due to its gradient boosting framework, regularization techniques, ability to handle missing values, efficient tree pruning, parallelization for speed, flexibility in objective functions and hyper parameters, feature importance analysis, and strong community support.

Decision Tree Regression Implementation: Decision tree regression is a machine learning algorithm that predicts the target variable by partitioning the feature space into smaller regions and fitting a simple model (usually a constant) within each region. In implementation, decision trees recursively split the data based on features to minimize variance in the target variable. The final prediction is made by averaging or taking the majority vote of the target values within the leaf nodes of the tree.

Random Forest Regression Implementation: Random forest regression is an ensemble learning technique that builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting. Each tree is trained on a random subset of the data and a random subset of features. In implementation, predictions are made by averaging the predictions of all the individual trees in the forest, providing robustness against noise and outliers.

Extra Trees Regression Implementation: Extra Trees, or Extremely Randomized Trees, is a variation of random forest where decision trees are trained on random subsets of the data and random thresholds for splitting, without the need for feature selection. In implementation, Extra Trees further randomizes the tree-building process by selecting random splitting thresholds for each feature at every node, leading to faster training and potentially higher accuracy, especially in high-dimensional datasets.

3.5 Evaluation

The models were evaluated based on the R2 score, with a particular focus on achieving an R2 score of around 0.5 as a benchmark for analysis.

These methodological choices pave the way for the following results, highlighting the predictive performance and insights gained.

4 Results and Discussion

4.1 Results

Table 1. Performance of Regression Models on Training Data

Model	R2 Score
Decision Tree Regressor	0.697
Random Forest Regressor	0.950
Extra Trees Regressor	0.911
XGBoost Regressor	0.876

Table 2. Performance of Regression Models on Test Data

Model	R2 Score
Decision Tree Regressor	0.718
Random Forest Regressor	0.815
Extra Trees Regressor	0.771
XGBoost Regressor	0.835

Within our investigation’s outcome analysis, we explored several machine learning models to predict the number of COVID-19 cases. The models implemented include Decision Tree Regression, Random Forest Regression, Extra Trees Regression, and the XGBoost Regression. Each of these models has inherent strengths and methodologies suitable for handling different data types and complexity.

When applied to the training dataset, these models yielded varying degrees of accuracy, with the Random Forest Regressor originally exhibiting the highest R2 score, an indication of its strong predictive performance.

However, the true test of a model’s generalizability and effectiveness comes when it is applied to unseen data. Upon evaluating these models on the test dataset, the XGBoost Regressor outperformed its counterparts, achieving the highest R2 score of 0.83510. This score not only reflects the model’s robustness but also its capability to capture the underlying patterns within the data, making it the most reliable model among those tested for forecasting COVID-19 case numbers in this project.

4.2 Discussion

In our study, the performance of various regression models, namely Decision Tree, Random Forest, Extra Trees, and XGBoost Regressors, exhibited variations in R2 scores when evaluated on training and test datasets. This difference in performance between the training and test phases can be attributed to several factors inherent in the model’s design, the nature of the data, and the unusual split of data for training and testing.

At first glance, the challenge presented by the dataset appears deceptively simple. However, its complexity defies traditional statistical and machine learning methods. The patterns within the dataset are exceptionally intricate, making them challenging to discern for researchers. Typically, removing multicollinearity is known to improve model performance, but such approaches prove ineffective with this dataset. The unconventional nature of the dataset’s structure introduces unique challenges, particularly in evaluating model performance.

Overfitting and Generalization Models like Decision Trees and Random Forests are prone to overfitting, especially when trained on smaller datasets. Overfitting occurs when a model learns the training data too well, including its noise and outliers, which reduces its ability to perform well on unseen data. The smaller training set size may have limited the models’ ability to generalize, leading to higher performance on the training data but diminished accuracy on the larger, more diverse test set.

Model Complexity The Random Forest and Extra Trees Regressors are ensemble methods that combine the predictions of several decision trees to improve predictive accuracy and control overfitting. The complexity of these models allows them to capture more nuances in the data compared to a single Decision Tree. However, the effectiveness of these ensemble methods is highly dependent

on the diversity and size of the training data. With a relatively smaller training set, the ensemble models might not have fully leveraged their complexity to generalize well to the extensive test dataset.

XGBoost’s Regularization The XGBoost Regressor incorporates built-in regularization terms to penalize complex models and reduce overfitting, making it more robust to variations in dataset sizes. This characteristic might explain why XGBoost exhibited a superior R2 score on the test data, as it could effectively balance between fitting the training data and generalizing it to the test data.

In conclusion, the discrepancies in R2 scores among the models between training and test datasets underscore the challenges of model selection and the importance of considering data distribution and size. We believe the findings highlight the need for careful consideration of model complexity, regularization techniques, and the balance between model fit and generalization, particularly in unconventional data split scenarios.